INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Alinged Data Center DFW-03, TX -- station KGKY, America/Chicago
Building	OSM 1443620406 Alinged Data Center DFW-03
Facade gap	not applicable -- one building, no second facade
Weather record	43,594 real hourly records from KGKY
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-01-21 (clear_cool)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 9 of 24 hours with 2 mode changes. 3 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	9 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	9 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 3 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	8.330	18.0	9.440	4.307
01	FREE	yes	-	8.880	18.0	10.000	3.891
02	FREE	yes	-	8.890	18.0	9.440	3.040
03	FREE	yes	-	9.440	18.0	10.000	3.090

04	FREE	yes	-	9.440	18.0	9.440	2.786
05	FREE	yes	-	9.440	18.0	9.440	2.543
06	FREE	yes	-	10.000	18.0	10.000	2.834
07	FREE	yes	-	11.670	18.0	10.560	2.853
08	FREE	yes	-	14.440	18.0	11.110	3.825
09	mechanical	no	dry-bulb	18.340	18.0	11.670	5.037
10	mechanical	no	dry-bulb	20.550	18.0	12.220	5.886
11	mechanical	no	dry-bulb	22.770	18.0	13.330	6.420
12	mechanical	no	dry-bulb	25.000	18.0	13.330	6.756
13	mechanical	no	dry-bulb	25.560	18.0	13.330	6.593
14	mechanical	no	dry-bulb	22.230	18.0	13.890	4.029
15	mechanical	no	dry-bulb	20.560	18.0	13.330	3.265
16	mechanical	no	dry-bulb	18.890	18.0	13.890	2.771
17	mechanical	yes	switch budget	17.780	18.0	13.890	2.679
18	mechanical	yes	switch budget	15.560	18.0	13.890	2.802
19	mechanical	yes	switch budget	13.880	18.0	13.890	3.355
20	mechanical	no	dew point	12.780	18.0	12.220	3.587
21	mechanical	no	dew point	11.660	18.0	11.110	4.185
22	mechanical	no	dew point	12.220	18.0	10.000	4.691
23	mechanical	no	dew point	12.220	18.0	8.890	4.474

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.330 C, which is 9.670 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.307 C, and the margin is measured, not chosen: 4.307 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.880 C, which is 9.120 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.891 C, and the margin is measured, not chosen: 3.891 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.890 C, which is 9.110 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.040 C, and the margin is measured, not chosen: 3.040 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.440 C, which is 8.560 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.090 C, and the margin is measured, not chosen: 3.090 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.440 C, which is 8.560 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.786 C, and the margin is measured, not chosen: 2.786 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.440 C, which is 8.560 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.543 C, and the margin is measured, not chosen: 2.543 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.000 C, which is 8.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.834 C, and the margin is measured, not chosen: 2.834 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.670 C, which is 6.330 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.853 C, and the margin is measured, not chosen: 2.853 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.440 C, which is 3.560 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.825 C, and the margin is measured, not chosen: 3.825 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.340 C against a 18.0 C limit, so it fails by 0.340 C. It would take a limit 0.340 C higher, or a bound 0.340 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.550 C against a 18.0 C limit, so it fails by 2.550 C. It would take a limit 2.550 C higher, or a bound 2.550 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.770 C against a 18.0 C limit, so it fails by 4.770 C. It would take a limit 4.770 C higher, or a bound 4.770 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.000 C against a 18.0 C limit, so it fails by 7.000 C. It would take a limit 7.000 C higher, or a bound 7.000 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.560 C against a 18.0 C limit, so it fails by 7.560 C. It would take a limit 7.560 C higher, or a bound 7.560 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.230 C against a 18.0 C limit, so it fails by 4.230 C. It would take a limit 4.230 C higher, or a bound 4.230 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.560 C against a 18.0 C limit, so it fails by 2.560 C. It would take a limit 2.560 C higher, or a bound 2.560 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.890 C against a 18.0 C limit, so it fails by 0.890 C. It would take a limit 0.890 C higher, or a bound 0.890 C tighter, to change this hour.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.780 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.560 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.880 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 12.780 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

21:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 11.660 C would have passed. The outside air is too HUMID: the dew-point bound is 15.05 C against a 15.0 C maximum, failing by 0.05 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

22:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 12.220 C would have passed. The outside air is too HUMID: the dew-point bound is 16.11 C against a 15.0 C maximum, failing by 1.11 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

23:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 12.220 C would have passed. The outside air is too HUMID: the dew-point bound is 16.11 C against a 15.0 C maximum, failing by 1.11 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

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